

Exploring Alpha Generation through Fixed Income Arbitrages

Preliminary Empirical Results — All Three Strategies

Alessio Ottaviani

EDHEC Business School

Supervisor: Prof. Riccardo Rebonato

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BTP Italia Basis

Returns are built from opening positions whenever the basis between BTP Italia and the matched nominal BTP becomes sufficiently wide (positive or negative). Positions remain open until convergence or maturity. Daily trade-level returns are aggregated into a single equal-weighted strategy index, resampled to monthly frequency.

CDS–Bond Basis

Universe includes all EUR corporate bonds whose issuer belongs to the iTraxx Main at the trade date. We only trade *negative* basis (long bond / long protection), for which implementation is liquid and feasible. Trades remain open until the basis reverts or the bond approaches maturity. Daily returns across all active trades form the CDS–Bond Basis index.

iTraxx Index Skew

We use the four EUR indices (Main, Senior Financials, Subordinated Financials, Crossover). At each decision date, only the on-the-run series are considered, each with thresholds calibrated to its volatility. The most mispriced index (in absolute basis) is traded. Returns are aggregated into a single iTraxx Skew index.

We evaluate our three fixed-income arbitrage strategies under the established risk-factor models used in the literature:

- *Duarte, Longstaff & Yu (2007), Review of Financial Studies.*
Fixed-income arbitrage benchmark used to assess exposure to equity, credit and Treasury-based relative-value factors.
- *Fung & Hsieh (2004), Financial Analysts Journal.*
Trend-following and option-based factors widely used to benchmark hedge-fund strategies.
- *Brooks, Gould & Richardson (2020), Journal of Fixed Income.*
Active fixed-income premium decomposition: term, credit, EM, inflation-linked, and volatility premia.

US factors are mapped to their EUR counterparts. European equity factors use the Fama-French Europe datasets, converted from USD into EUR following the methodology of Glück, Hübel & Scholz (JPM, 2020). Trend-following factors (Fung-Hsieh) are global and directly applicable.

Summary Statistics — Daily Returns (Annualized)

Strategy	n	Mean (% p.a.)	Std Dev (% p.a.)	Min (%)	Max (%)	Skew	Kurt	Ratio neg (%)	Serial corr	Sharpe
<i>BTP Italia</i>	3282	1.74	2.49	-1.59	1.80	0.60	19.94	49.3	-0.045	0.70
<i>CDS Bond Basis</i>	4173	6.71	3.25	-1.56	4.30	4.45	81.74	42.1	-0.056	2.06
<i>iTraxx Combined</i>	4883	4.27	3.11	-0.95	1.60	0.62	5.44	47.0	-0.205	1.38

Notes: This table reports summary statistics for daily percentage returns. *n* is the number of daily observations. *Mean* and *Std Dev* are annualized (252 trading days). *Skew* and *Kurt* denote skewness and excess kurtosis. *Ratio neg* is the proportion of negative returns. *Serial corr* is the first-order autocorrelation coefficient. *Sharpe* is annualized. Sample period varies by strategy.

SW vs EW — Basic Metrics (Daily, Annualized)

Strategy	Return (% p.a.)	Vol (% p.a.)	Sharpe	MaxDD (%)	Skew	Kurt	$\Theta_{\rho=3}$ (% p.a.)	N
Panel A: EW								
BTP-Italia	1.74	2.49	0.70	-2.97	0.60	19.94	1.65	159
iTraxx <i>combined</i>	4.27	3.12	1.37	-2.04	0.62	5.41	4.12	242
CDS-Bond Basis	6.71	3.25	2.06	-4.05	4.45	81.74	6.55	203
Panel B: SW								
BTP-Italia	1.99	2.38	0.84	-2.97	0.73	23.33	1.90	159
iTraxx <i>combined</i>	4.85	3.39	1.43	-2.52	0.64	5.26	4.68	242
CDS-Bond Basis	7.55	3.55	2.13	-3.47	4.48	68.94	7.37	203

Note: Metrics computed from daily returns (annualization factor 252). MaxDD is computed on the wealth index. $\Theta_{\rho=3}$ is the Goetzmann et al. (2007) manipulation-proof measure.

SW vs EW — MPPM Comparison (Monthly)

Strategy	$\rho = 2$ (% p.a.)	$\rho = 3$ (% p.a.)	$\rho = 4$ (% p.a.)	Avg Ret (% p.a.)	Vol (% p.a.)	Sharpe	N
Panel A: EW							
BTP-Italia	1.62	1.61	1.59	1.65	1.77	0.93	163
iTraxx <i>combined</i>	3.98	3.95	3.93	4.03	2.10	1.92	243
CDS-Bond Basis	6.32	6.26	6.20	6.47	3.59	1.80	207
Panel B: SW							
BTP-Italia	1.86	1.85	1.83	1.89	1.71	1.11	163
iTraxx <i>combined</i>	4.51	4.49	4.46	4.58	2.39	1.91	243
CDS-Bond Basis	7.12	7.05	6.98	7.29	3.89	1.87	207

Note: MPPM is the Manipulation-Proof Performance Measure (Goetzmann et al. 2007). Computed from monthly returns compounded from daily. Risk-free is Euribor 1M (ACT/360), compounded to monthly.

SW vs EW — Volatility-Managed Performance (Monthly)

Strategy	Buy-and-Hold			Vol-Managed			Alpha	N
	Ret (% p.a.)	Vol (% p.a.)	Sharpe	Ret (% p.a.)	Vol (% p.a.)	Sharpe		
Panel A: EW								
BTP-Italia	1.57	1.74	0.90	0.41	1.74	0.23	-0.28	158
iTraxx <i>combined</i>	4.05	2.10	1.93	1.75	2.10	0.83	+0.15	241
CDS-Bond Basis	6.38	3.55	1.80	4.54	3.55	1.28	+2.79	202
Panel B: SW								
BTP-Italia	1.82	1.68	1.08	0.55	1.68	0.33	-0.14	158
iTraxx <i>combined</i>	4.62	2.40	1.93	1.85	2.40	0.77	-0.06	241
CDS-Bond Basis	7.22	3.86	1.87	5.44	3.86	1.41	+3.45	202

Note: Following Moreira and Muir (2017). The volatility-managed strategy scales positions inversely to realized variance: $f_{t+1}^{\sigma} = (c/\sigma_t^2) \times f_{t+1}$, where c matches buy-and-hold volatility. Alpha is from regressing VM returns on BH returns.

SW vs EW — Volatility-Managed Robustness (Monthly)

Strategy	Metric	No Leverage (≤ 1)	50% Leverage (≤ 1.5)	Table V [0.1, 2.0]
Panel A: EW				
BTP-Italia	Alpha (% p.a.)	+0.16	+0.17	+0.16
	Sharpe	0.84	0.79	0.79
iTraxx <i>combined</i>	Alpha (% p.a.)	+0.11	+0.11	+0.10
	Sharpe	1.27	1.11	1.09
CDS-Bond Basis	Alpha (% p.a.)	+0.80	+1.05	+1.15
	Sharpe	1.43	1.42	1.50
Panel B: SW				
BTP-Italia	Alpha (% p.a.)	+0.11	+0.14	+0.14
	Sharpe	0.90	0.89	0.91
iTraxx <i>combined</i>	Alpha (% p.a.)	+0.01	-0.08	-0.15
	Sharpe	1.10	0.94	0.92
CDS-Bond Basis	Alpha (% p.a.)	+1.00	+1.46	+1.64
	Sharpe	1.49	1.50	1.60

Note: Robustness tests following Table V in Moreira and Muir (2017). *No Leverage:* scaling $\in [0, 1]$. *50% Table V:* scaling $\in [0.1, 2.0]$ as in the paper.

Duarte Factor Model

	α (%)	β_{Mkt-RF}	β_{SMB}	β_{HML}	β_{UMD}	β_{RS}	β_{RI}	β_{RB}	β_{R2}	β_{R5}	β_{R10}	R^2 adj	N
Panel A: US Factors													
<i>BTP Italia</i>	2.12*** (3.97)	-0.063*** (-3.10)	-0.026* (-1.87)	-0.042** (-2.12)	-0.003 (-0.19)	0.032*** (2.74)	0.012 (0.14)	-0.067 (-0.72)	0.166 (0.65)	-0.112 (-0.63)	0.080 (0.97)	0.108	161
<i>CDS Bond Basis</i>	4.95*** (6.10)	-0.041* (-1.88)	0.006 (0.34)	0.000 (0.01)	-0.026* (-1.67)	0.010 (0.77)	0.217** (2.52)	-0.068 (-1.63)	-0.309 (-0.74)	-0.013 (-0.06)	0.025 (0.26)	0.138	204
<i>iTraxx Indices Skew</i>	3.35*** (6.47)	0.022 (1.34)	-0.013 (-1.13)	-0.020 (-1.53)	-0.027*** (-2.74)	-0.013 (-1.47)	-0.002 (-0.04)	-0.036 (-1.12)	0.351 (1.46)	-0.061 (-0.39)	0.014 (0.20)	0.064	240
Panel B: EUR Factors													
<i>BTP Italia</i>	1.63*** (2.98)	-0.048* (-1.78)	-0.058** (-2.33)	-0.054** (-2.08)	0.030 (1.34)	0.024* (1.67)	0.008 (0.08)	-0.032 (-0.34)	0.055 (0.11)	-0.021 (-0.07)	-0.029 (-0.24)	0.112	162
<i>CDS Bond Basis</i>	5.03*** (5.11)	-0.031 (-1.36)	-0.004 (-0.14)	0.005 (0.16)	-0.048* (-1.77)	-0.015 (-1.37)	0.380*** (2.87)	-0.069 (-1.56)	-0.711 (-1.21)	0.609 (1.61)	-0.271** (-2.09)	0.171	205
<i>iTraxx Indices Skew</i>	3.83*** (6.83)	-0.013 (-0.73)	-0.058** (-2.40)	-0.047* (-1.87)	-0.040*** (-2.67)	-0.007 (-0.56)	0.129** (2.32)	-0.027 (-1.06)	0.017 (0.03)	0.176 (0.52)	-0.159 (-1.34)	0.104	241

Monthly excess returns regressed on the indicated equity and bond factors. Alpha in annualized percentage terms. Newey-West HAC t-statistics in parentheses. Mkt-RF, SMB, HML, and UMD are the Fama-French market, small-minus-big, high-minus-low, and up-minus-down (momentum) factors. R_S is the excess return on an S&P index of bank stocks. R_I , R_B are the excess returns on Merrill Lynch indexes of A/BAA-rated industrial bonds and A/BAA-rated bank bonds, respectively. R_2 , R_5 , and R_{10} are the excess returns on maturity-sorted bond portfolios of 2-year, 5-year, and 10-year Treasury bonds, respectively. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

$$R_{it} = \alpha + \beta_1 R_{Mkt} + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 UMD_t + \beta_5 R_{S,t} + \beta_6 R_{I,t} + \beta_7 R_{B,t} + \beta_8 R_{2,t} + \beta_9 R_{5,t} + \beta_{10} R_{10,t} + \varepsilon_t$$

Fung & Hsieh Factor Model

	α (%)	$\beta_{S\&P}$	β_{SC-LC}	β_{BdOpt}	β_{FXOpt}	β_{ComOpt}	β_{10Y}	$\beta_{CredSpr}$	R^2 adj	N
Panel A: US Factors										
<i>BTP Italia</i>	1.79*** (3.26)	-0.017* (-1.75)	-0.025* (-1.93)	0.264 (1.36)	-0.182 (-0.84)	0.292 (0.84)	0.015 (0.07)	0.422 (1.63)	0.103	162
<i>CDS Bond Basis</i>	5.19*** (6.22)	-0.027** (-2.34)	0.015 (0.95)	0.111 (0.38)	-0.381 (-1.14)	-0.375 (-1.06)	-1.045*** (-3.68)	-1.108*** (-2.74)	0.138	205
<i>iTraxx Indices Skew</i>	3.27*** (6.17)	0.015 (1.17)	-0.015 (-1.12)	0.101 (0.47)	-0.040 (-0.18)	0.460* (1.70)	-0.125 (-0.79)	0.224 (0.88)	0.018	241
Panel B: EUR Factors										
<i>BTP Italia</i>	1.86*** (3.64)	-0.039*** (-2.99)	-0.059** (-2.40)	0.341* (1.80)	-0.082 (-0.39)	0.337 (0.92)	0.284 (1.44)	-0.530 (-1.13)	0.111	162
<i>CDS Bond Basis</i>	4.67*** (5.38)	-0.029 (-1.60)	0.040 (1.22)	0.209 (0.88)	-0.709* (-1.80)	-0.593 (-1.64)	-0.884** (-2.38)	-0.735 (-1.32)	0.112	195
<i>iTraxx Indices Skew</i>	3.48*** (6.05)	-0.004 (-0.28)	-0.065** (-2.02)	0.118 (0.53)	-0.130 (-0.56)	0.587* (1.83)	-0.336 (-1.37)	-0.123 (-0.38)	0.051	195

Monthly excess returns regressed on equity, spread and trend-following factors. Alpha in annualized percentage terms. Newey-West HAC t-statistics in parentheses. S&P is the S&P 500 index. SC-LC is the Wilshire small-minus-large equity return. 10Y is the change in the 10y constant-maturity yield. CredSpr is the change in Moody's Baa - 10y Treasury spread. BdOpt, FXOpt, ComOpt are lookback-straddle portfolios on bond, FX and commodity futures.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

$$R_{it} = \alpha + \beta_1 S\&P_t + \beta_2 SC-LC_t + \beta_3 BdOpt_t + \beta_4 FXOpt_t + \beta_5 ComOpt_t + \beta_6 10Y_t + \beta_7 CredSpr_t + \varepsilon_t$$

Active Fixed Income Factor Model

	α (%)	β_{Term}	$\beta_{GlobalTerm}$	$\beta_{GlobalAgg}$	$\beta_{Infl.Linkers}$	$\beta_{CorpCredit}$	$\beta_{EmergDebt}$	$\beta_{EmergCurr}$	β_{USTVol}	R^2	adj	N
Panel A: US Factors												
<i>BTP Italia</i>	2.10*** (3.88)	0.304* (1.73)	-0.264 (-0.83)	-0.076 (-0.18)	-0.022 (-0.58)	-0.133** (-2.03)	0.104 (1.54)	-0.024 (-0.70)	-0.002 (-0.19)	0.145		163
<i>CDS Bond Basis</i>	4.24*** (5.56)	0.016 (0.11)	-0.335 (-0.82)	0.645 (1.31)	-0.069 (-1.28)	-0.009 (-0.17)	0.035 (0.68)	-0.023 (-0.77)	0.008 (0.43)	0.097		207
<i>iTraxx Indices Skew</i>	3.04*** (6.49)	0.060 (0.49)	0.401* (1.69)	-0.290 (-1.04)	-0.061 (-1.57)	0.060 (1.40)	-0.018 (-0.42)	0.029 (1.10)	0.032** (2.37)	0.062		243
Panel B: EUR Factors												
<i>BTP Italia</i>	2.19*** (3.07)	-0.198*** (-3.32)	-0.450 (-1.62)	0.736** (2.32)	-0.069 (-1.62)	-0.086 (-1.39)	-0.068 (-1.38)	0.056 (1.55)	-0.001 (-0.09)	0.052		163
<i>CDS Bond Basis</i>	4.79*** (5.83)	0.161 (1.61)	-0.124 (-0.47)	0.217 (0.92)	-0.061 (-1.18)	-0.068 (-1.16)	0.098* (1.76)	-0.043 (-1.00)	-0.010 (-1.01)	0.063		207
<i>iTraxx Indices Skew</i>	3.34*** (6.58)	-0.048 (-0.49)	0.260 (1.22)	-0.039 (-0.18)	-0.044 (-1.20)	0.011 (0.44)	-0.002 (-0.08)	0.006 (0.28)	-0.014 (-1.36)	0.008		243

Monthly excess returns regressed on traditional fixed income premia (term, credit, EM, inflation-linked) and volatility. Alpha in annualized percentage terms. Newey-West HAC t-statistics in parentheses. Term: Bloomberg Barclays Treasury excess returns (US or Eurozone). Global Term/Agg: BBG Global Treasury/Aggregate (hedged). Inflation-Linkers: BBG Global Inflation-Linked. Corporate Credit: HY + leveraged loans in excess of duration-matched Treasuries. Emerging Debt/Currency: EM bond and FX baskets. UST Vol: delta-hedged straddles on 10y Treasury futures.

*** p<0.01, ** p<0.05, * p<0.10.

$$R_{it} = \alpha + \beta_1 R_{Term,t} + \beta_2 R_{GlobalTerm,t} + \beta_3 R_{GlobalAgg,t} + \beta_4 R_{InflLinkers,t} + \beta_5 R_{CorpCredit,t} + \beta_6 R_{EmergDebt,t} + \beta_7 R_{EmergCurr,t} + \beta_8 \Delta USTVol_t + \varepsilon_t$$

- **Factor universe:** PCA on the full set of 70+ factors (tradable, non-tradable, macro and transformed variables); no grouping by factor families.
- **Estimation scheme:** out-of-sample rolling PCA with fixed window W (baseline $W = 24$ months); loadings estimated on $[t - W, t - 1]$.
- **Timing:** both contemporaneous (R_t on PC_t) and predictive (R_{t+1} on PC_t) specifications.
- **Inference:** Newey–West HAC standard errors; robustness to window length W and number of PCs K .

PCA Spanning Regression — Alpha

Strategy	Predictive ($PC_t \rightarrow R_{t+1}$)			Contemporaneous ($PC_t \rightarrow R_t$)		
	α (%)	R^2 adj	N	α (%)	R^2 adj	N
<i>Btp Italia</i>	1.47*** (2.73)	0.036	157	1.37** (2.35)	0.040	158
<i>Cds Bond Basis</i>	6.38*** (4.43)	-0.013	198	6.53*** (4.82)	0.073	199
<i>Itraxx Combined</i>	3.31*** (4.19)	0.002	181	2.61*** (3.92)	0.103	182

Notes: Alpha is annualized. PC1–PC8 are the first 8 principal components extracted from 24-month rolling windows. The first 8 PCs explain on average 79.2% of factor variance (min 73.1%, max 86.6%). t-statistics in parentheses based on Newey-West HAC standard errors.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

PCA Spanning Regression — Full Model

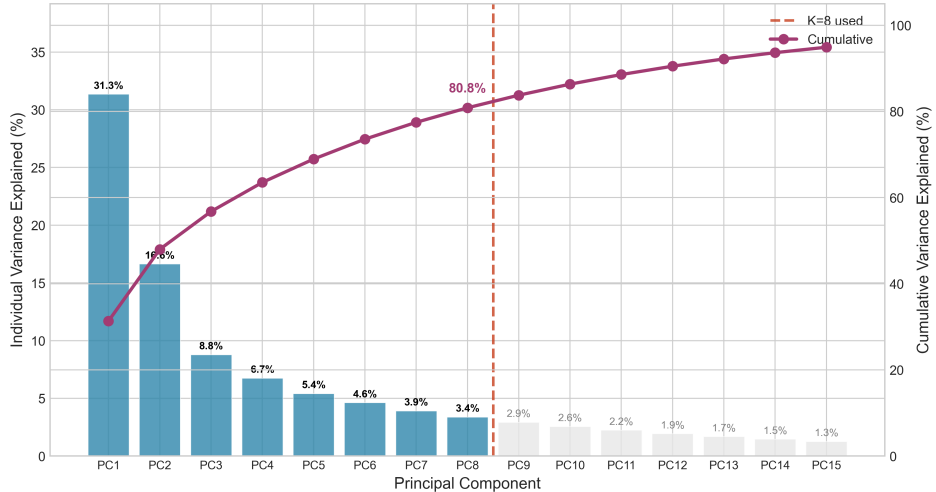
Strategy	α (%)	β_{PC1}	β_{PC2}	β_{PC3}	β_{PC4}	β_{PC5}	β_{PC6}	β_{PC7}	β_{PC8}	R^2 adj	N
Panel A: Predictive ($PC_t \rightarrow R_{t+1}$)											
<i>Btp Italia</i>	1.47*** (2.73)	0.006 (0.45)	0.011 (0.78)	-0.022 (-1.12)	-0.026 (-0.71)	-0.058*** (-2.80)	0.016 (0.55)	-0.011 (-0.36)	-0.073** (-2.04)	0.036	157
<i>Itraxx Combined</i>	3.31*** (4.19)	-0.004 (-0.26)	0.017 (0.80)	-0.054** (-2.03)	-0.021 (-0.76)	0.013 (0.48)	-0.027 (-0.78)	0.066** (2.27)	-0.002 (-0.08)	0.002	181
<i>Cds Bond Basis</i>	6.38*** (4.43)	-0.006 (-0.27)	-0.037 (-1.55)	-0.038 (-1.00)	0.015 (0.30)	-0.011 (-0.33)	0.035 (1.03)	0.030 (0.49)	0.025 (0.53)	-0.013	198
Panel B: Contemporaneous ($PC_t \rightarrow R_t$)											
<i>Btp Italia</i>	1.37** (2.35)	0.004 (0.39)	0.029 (1.62)	0.013 (0.47)	0.012 (0.46)	0.027 (1.03)	0.019 (0.82)	0.047* (1.75)	0.032 (1.29)	0.040	158
<i>Itraxx Combined</i>	2.61*** (3.92)	-0.022** (-2.24)	0.005 (0.22)	0.004 (0.15)	0.021 (0.60)	-0.008 (-0.27)	0.073*** (2.64)	0.072*** (2.68)	-0.011 (-0.53)	0.103	182
<i>Cds Bond Basis</i>	6.53*** (4.82)	-0.024 (-0.96)	-0.043 (-1.35)	0.059 (1.11)	-0.099*** (-3.04)	0.006 (0.10)	-0.081** (-2.05)	0.023 (0.47)	0.070 (1.42)	0.073	199

This table reports spanning regression results of monthly strategy excess returns on the first 8 principal components. Principal components are extracted from a rolling window of 24 months. Alpha is reported in annualized percentage terms. t-statistics in parentheses are based on Newey-West HAC standard errors. The first 8 PCs explain on average 79.2% of factor variance (min 73.1%, max 86.6%).

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

PCA Scree Plot

PCA Variance Explained by Principal Component
(Rolling Window: 24 months, K=8 used in regressions)



PCA Robustness — Window Length and Number of PCs

Robustness (W,K): Alpha (t) and Adjusted R^2 — Predictive vs Contemporaneous

W	K	Btp Italia					Cds Bond Basis					Itraxx Combined					Expl. Var. (%)
		Pred		Cont		R^2_{adj}	Pred		Cont		R^2_{adj}	Pred		Cont		R^2_{adj}	
		α	R^2_{adj}	α	R^2_{adj}		α	R^2_{adj}	α	R^2_{adj}		α	R^2_{adj}	α	R^2_{adj}		
24	8	1.51*** (3.38)	0.020	1.63*** (3.22)	0.011	6.54*** (4.83)	0.001	7.40*** (4.87)	0.065	4.52*** (5.84)	0.017	4.63*** (6.19)	0.003	80.8			
24	11	1.56*** (3.34)	0.026	1.69*** (3.16)	0.007	6.34*** (4.69)	0.002	7.53*** (4.75)	0.062	4.55*** (5.84)	0.003	4.48*** (6.18)	0.051	88.6			
24	14	1.47*** (2.94)	0.073	1.80*** (3.11)	0.028	6.31*** (4.72)	0.020	7.37*** (4.81)	0.059	4.61*** (5.70)	-0.006	4.46*** (6.20)	0.042	93.6			
36	8	1.59*** (3.23)	0.061	1.51*** (3.32)	0.081	7.03*** (5.07)	0.051	6.72*** (4.63)	0.014	4.62*** (5.80)	0.059	4.22*** (5.20)	0.072	76.2			
36	11	1.59*** (3.38)	0.043	1.43*** (3.35)	0.090	7.25*** (4.96)	0.076	7.08*** (4.78)	0.037	4.58*** (5.61)	0.068	4.31*** (5.19)	0.065	83.6			
36	14	1.50*** (3.10)	0.044	1.41*** (3.41)	0.083	7.24*** (5.06)	0.061	7.07*** (4.84)	0.039	4.63*** (5.68)	0.071	4.38*** (5.23)	0.062	88.8			
48	8	2.01*** (4.41)	0.129	1.68*** (3.67)	0.052	6.20*** (4.02)	0.008	6.69*** (4.43)	0.038	4.59*** (5.55)	0.114	4.00*** (5.01)	0.122	73.2			
48	11	1.99*** (4.80)	0.143	1.64*** (3.59)	0.048	6.01*** (4.08)	0.043	6.66*** (4.66)	0.026	4.45*** (5.60)	0.117	4.04*** (5.15)	0.118	80.5			
48	14	1.87*** (4.04)	0.145	1.77*** (4.00)	0.046	6.04*** (4.35)	0.035	6.70*** (4.24)	0.026	4.53*** (5.51)	0.107	4.12*** (5.27)	0.119	85.8			

W is the rolling window length (months) used to estimate PCA factors; K is the number of retained principal components. Expl. Var. (%) is the average cumulative variance explained by the first K PCs (rolling, time-series average). Alpha is annualized (% p.a.). t-statistics in parentheses (Newey-West HAC). Pred = $PC_t \rightarrow R_{t+1}$, Cont = $PC_t \rightarrow R_t$. *** p<0.01, ** p<0.05, * p<0.10.

MPPM and Volatility-Managed Performance

Manipulation-Proof Performance Measure (MPPM)

Strategy	MPPM (% p.a.)			Avg Return (% p.a.)	Vol Sharpe (% p.a.)	N
	$\rho = 2$	$\rho = 3$	$\rho = 4$			
CDS-Bond Basis	4.86	4.82	4.79	4.93	2.62	1.89 207
iTraxx <i>combined</i>	3.38	3.37	3.35	3.42	1.84	1.86 243
BTP-Italia	1.58	1.57	1.55	1.62	1.84	0.88 163

MPPM as in Goetzmann et al. (2007), based on monthly returns and reported in annualized percentage terms. All moments are annualized.

Volatility-Managed Portfolios (Moreira-Muir, 2017)

Strategy	Buy-and-Hold			Volatility-Managed			Regression			N
	Return (% p.a.)	Vol (% p.a.)	Sharpe	Return (% p.a.)	Vol (% p.a.)	Sharpe	Alpha (% p.a.)	Beta	R^2	
CDS-Bond Basis	4.92	2.62	1.88	3.13	2.62	1.19	+2.00	0.23	0.053	202
iTraxx <i>combined</i>	3.43	1.84	1.86	1.63	1.84	0.89	-0.04	0.49	0.239	242
BTP-Italia	1.50	1.80	0.84	1.08	1.80	0.60	+0.10	0.65	0.422	162

Volatility-managed portfolios scale exposure inversely with realized variance so that unconditional volatility matches buy-and-hold. Regression alpha measures the volatility-timing premium from $r_{t+1}^{VM} = \alpha + \beta r_{t+1} + \varepsilon_{t+1}$.

- All three strategies deliver positive MPPM, with the highest utility-based performance for the CDS-Bond basis and iTraxx strategies.
- Volatility timing generates a sizable premium for the CDS-Bond basis, while effects are more modest for BTP Italia and iTraxx, consistent with their lower return variance.

Volatility Timing Robustness — Leverage Constraints

Strategy	Metric	No Leverage (≤ 1)	50% Leverage (≤ 1.5)	Table V [0.1, 2.0]
CDS-Bond Basis	Alpha (% p.a.)	+0.60	+0.80	+0.83
	Sharpe	1.53	1.48	1.49
iTraxx <i>combined</i>	Alpha (% p.a.)	+0.03	+0.01	-0.02
	Sharpe	1.35	1.18	1.09
BTP-Italia	Alpha (% p.a.)	+0.02	+0.14	+0.21
	Sharpe	0.69	0.73	0.74

Leverage constraints follow Moreira and Muir (2017), Table V. “No Leverage” constrains the scaling factor to $[0, 1]$, “50% Leverage” to $[0, 1.5]$, and “Table V” to $[0.1, 2.0]$. Alpha measures the volatility-timing premium under each constraint.

- Volatility-timing alpha for the CDS–Bond basis remains positive and stable across all leverage constraints, indicating that results are not driven by extreme leverage.
- For BTP Italia and iTraxx, volatility timing adds a smaller but consistent premium, with Sharpe ratios largely preserved under realistic constraints.

What has been done since then (new work)

To strengthen the risk-adjustment side beyond the standard benchmarks, I conducted an extensive review of the recent fixed-income and arbitrage literature and **reconstructed the full historical time-series dataset** for a comprehensive set of candidate risk controls (70+ factors). Each series was rebuilt consistently with the original paper definitions (transformations, shocks/innovations, lags, and frequency alignment).

Factor universe (organized in 6 families)

- *Credit risk* (e.g., CDS indices, excess bond premium, sovereign spreads)
- *Liquidity & funding stress* (e.g., OIS spreads, repo indicators, noise/illiquidity measures)
- *Equity / intermediary proxies* (e.g., BAB, value, momentum)
- *Volatility* (e.g., VIX/V2X, MOVE, option-implied measures, trend-following)
- *Macro uncertainty & policy* (e.g., uncertainty indices, EPU, financial conditions)
- *Rates & term structure* (e.g., term, slope, swap spreads)

Next: full factor list (as a “menu”) and then a parsimonious spanning set will be identified for each strategy.



Risk Factors Universe (Literature-Based, Reconstructed Series) I

Factor ID	Description	Reference
Panel A: Credit Risk Factors		
BTP_BUND	Yield differential between the Italian 10-year BTP and the German Bund.	Fausch and Sigonius (2018)
CDX_IG	First difference of the CDS spread on the CDX Investment Grade index.	Klaus and Rzepkowski (2009)
CREDIT_EU	Same construction as CREDIT_US, computed using Bloomberg euro-area corporate bond yield indices for BBB and AAA ratings.	Fama and French (1989)
CREDIT_US	Bond credit risk factor. Yield spread between Moody's BAA and AAA corporate bond indices.	Fama and French (1989)
DEF_EU	Same construction as DEF_US, computed using long-term European corporate bond total returns relative to long-term German Bund total returns.	Fama and French (1993)

Risk Factors Universe (Literature-Based, Reconstructed Series) II

DEF_US	Bond default risk factor. Total return spread between a long-term U.S. corporate bond portfolio and a long-term U.S. government bond portfolio.	Fama and French (1993)
EBP	Excess Bond Premium. Residual component of aggregate corporate bond spreads after netting out expected default risk and bond characteristics via cross-sectional spread regressions; proxies time-varying credit risk compensation tied to intermediary constraints. Used with a one-month lag.	Gilchrist and Zakrajšek (2012)
ITRX_MAIN	First difference of the CDS spread on the iTraxx Europe Main index.	Klaus and Rzepkowski (2009)
MAIN_5/3_FLATTENED	First difference of the natural logarithm of the ratio between the 5-year and 3-year iTraxx Europe Main CDS spreads.	Market indicator
PB_EU_CDS_1Y	Average 1-year CDS spread of European prime brokers.	Klaus and Rzepkowski (2009)
PB_EU_CDS_5Y	Average 5-year CDS spread of European prime brokers.	Klaus and Rzepkowski (2009)
PB_US_CDS_1Y	Average 1-year CDS spread of U.S. prime brokers.	Klaus and Rzepkowski (2009)

Risk Factors Universe (Literature-Based, Reconstructed Series) III

PB_US_CDS_5Y	Average 5-year CDS spread of U.S. prime brokers.	Klaus and Rzepkowski (2009)
SNR-MAIN	First difference of the 5-year CDS spread difference between the iTraxx Senior Financials index and the iTraxx Europe Main index.	Market indicator
XOVER/MAIN	First difference of the natural logarithm of the ratio between the 5-year iTraxx Crossover and 5-year iTraxx Europe Main CDS spreads.	Market indicator

Panel B: Liquidity Factors

Δ FAILS_PCT_TSY	First difference of the ratio of Monthly total notional amount of U.S. Treasury securities fails-to-deliver reported by primary dealers, excluding TIPS, to U.S. Total Debt Outstanding, expressing repo fails as a share of total marketable U.S. Treasury debt.	Fleckenstein et al. (2014)
EURIBOR_OIS	Euribor–OIS spread. Difference between 3-month Euribor and the 3-month overnight index swap rate based on EONIA or €STR.	Nyborg and Östberg (2014)
GC-REPO_T-BILL	Spread between the 3-month U.S. Treasury general-collateral repo rate and the 3-month U.S. Treasury bill yield.	Bai and Collin-Dufresne (2019)

Risk Factors Universe (Literature-Based, Reconstructed Series) IV

HKM_IC	Intermediary capital factor. AR(1) innovation in the market-based capital ratio of primary dealers, scaled by the lagged capital ratio.	He et al. (2017)
HPW_NOISE	Treasury noise measure of market-wide illiquidity. Root mean squared deviation between observed U.S. Treasury yields and model-implied yields from a fitted smooth zero-coupon yield curve, aggregated across bonds with maturities between 1 and 10 years.	Hu et al. (2013)
ILLIQ	Change in the Amihud illiquidity measure. First difference of the ratio of absolute daily return to daily dollar volume, capturing shocks to market illiquidity.	Amihud and Mendelson (2015)
LIBOR_OIS	Libor–OIS spread. Difference between 3-month LIBOR and the 3-month overnight index swap rate.	Nyborg and Östberg (2014)
LIBOR_REPO_SHOCK	Defined as the AR(2) residual of the spread, where the spread is the monthly average of daily 3-month U.S. interbank LIBOR minus the 3-month U.S. Treasury general-collateral repo rate.	Asness et al. (2013)
LIQ_V	Value-weighted return on a 10–1 portfolio formed by sorting stocks on historical liquidity betas.	Pástor and Stambaugh (2003)
LIQNT	Pastor–Stambaugh non-traded liquidity factor. Innovation in aggregate market liquidity constructed from a cross-sectional average of stock-level order-flow-induced return-reversal measures, with predictable components removed.	Pástor and Stambaugh (2003)

Risk Factors Universe (Literature-Based, Reconstructed Series) V

SILLIQ	Stock illiquidity shock. Innovation in the monthly stock-market illiquidity series, defined as the AR(3) residual of the aggregate Amihud illiquidity measure.	Acharya et al. (2013)
TED_SHOCK_EU	Same construction as TED_SHOCK_US, computed using the monthly average of daily 3-month Euribor minus the 3-month German government bill rate.	Asness et al. (2013)
TED_SHOCK_US	Defined as the AR(2) residual of the TED spread, where the TED spread is the monthly average of daily 3-month U.S. interbank LIBOR minus the 3-month U.S. government bill rate.	Asness et al. (2013)

Panel C: Equity Factors

BAB_EU	Betting Against Beta (Europe). Same construction as BAB_US, implemented on the European equity universe.	Frazzini and Pedersen (2014)
BAB_US	Betting Against Beta (US). Long low-beta stocks and short high-beta stocks; each leg is rescaled to unit beta, forming a near market-neutral zero-cost factor.	Frazzini and Pedersen (2014)
CMA_EU	Investment factor (Conservative Minus Aggressive). Return spread between European conservative and aggressive investment portfolios from independent size-investment sorts.	Fama and French (2017)

Risk Factors Universe (Literature-Based, Reconstructed Series) VI

GMOM_EU	Global momentum factor. Return spread between recent winner and loser equity portfolios based on past 12-month returns excluding the most recent month, expressed in euros.	Asness et al. (2013)
GVAL_EU	Global value factor. Return spread between high and low book-to-market equity portfolios formed within major developed equity regions, expressed in euros.	Asness et al. (2013)
HML_EU	Value factor (High Minus Low). Return spread between European high and low book-to-market portfolios from independent size-B/M sorts, approximately size-neutral.	Fama and French (2017)
MKT_EU	Market excess return on the European value-weighted equity market portfolio over the 1-month German government bill. Returns are expressed in euros.	Fama and French (2017)
RMW_EU	Profitability factor (Robust Minus Weak). Return spread between European robust and weak operating profitability portfolios from independent size-profitability sorts.	Fama and French (2017)
SMB_EU	Size factor (Small Minus Big). Return spread between diversified European small- and large-cap portfolios from independent size sorts (five-factor construction).	Fama and French (2017)
UMD_EU	Momentum factor (Up Minus Down). Monthly return on a European zero-investment winners-minus-losers momentum portfolio.	Carhart (1997)

Risk Factors Universe (Literature-Based, Reconstructed Series) VII

Panel D: Volatility Factors

$\Delta V2X$	Change in V2X. Monthly first difference of the V2X index, the European counterpart of VIX measuring 30-day implied volatility on Euro STOXX 50 options.	Chung et al. (2019)
ΔVIX	Change in VIX. Monthly first difference of the VIX index, a model-free measure of 30-day implied volatility on S&P 500 options.	Chung et al. (2019)
ATM_IV_CDX	3-month at-the-money implied volatility on options written on the CDX Investment Grade index.	Novales and Chamizo (2019)
ATM_IV_ITRX	3-month at-the-money implied volatility on options written on the iTraxx Europe Main index.	Novales and Chamizo (2019)
EP_SVIX_1M	Equity premium proxy from the SVIX methodology. Option-implied risk-neutral variance of the market simple return over the next month, computed by integrating out-of-the-money index option prices across strikes.	Martin (2017)
EP_SVIX_3M	Same construction as EP_SVIX_1M, using options with a 3-month horizon.	Martin (2017)
IV_BUND	3-month at-the-money implied volatility on the 10-year German Bund futures options, based on 100% moneyness.	Cremers et al. (2021)

Risk Factors Universe (Literature-Based, Reconstructed Series) VIII

IV_TSY	3-month at-the-money implied volatility on the 10-year U.S. Treasury futures options, based on 100% moneyness.	Cremers et al. (2021)
MOVE	Option-implied U.S. rates volatility. Yield-curve weighted average of at-the-money implied normal yield volatilities from OTC options on 2-, 5-, 10-, and 30-year constant-maturity interest rate swaps; used lagged by one month.	Bansal and Stivers (2022)
MOVE2	Squared MOVE; used lagged by one month to capture nonlinear exposure to option-implied U.S. interest rate volatility.	Bansal and Stivers (2022)
PTFSBD	Return on the bond trend-following factor, constructed from lookback straddles on government bond futures.	Fung and Hsieh (2001)
PTFSCOM	Return on the commodity trend-following factor, constructed from lookback straddles on commodity futures.	Fung and Hsieh (2001)
PTFSFX	Return on the currency trend-following factor, constructed from lookback straddles on major currency futures.	Fung and Hsieh (2001)
PTFSIR	Return on the short-term interest rate trend-following factor, constructed from lookback straddles on 3-month interest rate futures.	Fung and Hsieh (2001)

Risk Factors Universe (Literature-Based, Reconstructed Series) IX

PTFSSTK	Return on the equity index trend-following factor, constructed from lookback straddles on stock index futures.	Fung and Hsieh (2001)
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Panel E: Macro Factors

$\Delta U F$	Change in financial uncertainty. First difference of the financial uncertainty index, constructed as $\Delta U M$ but using a panel of financial market variables.	Ludvigson et al. (2021)
$\Delta U M$	Change in macroeconomic uncertainty. First difference of the macro uncertainty index, a model-based measure of expected volatility in the unforecastable component of a large panel of macroeconomic variables.	Ludvigson et al. (2021)
$\Delta U R$	Change in real uncertainty. First difference of the real-activity uncertainty subindex, constructed from real activity variables.	Ludvigson et al. (2021)
5Y5YREAL_YIELD	First difference of the spread between the 5y5y interest rate swap rate and the 5y5y inflation swap rate.	Market indicator
BFCI_EU	Bloomberg Euro Area Financial Conditions Index. Composite Z-score of euro-area money, bond, and equity indicators, standardized to the 1999 to July 1, 2008 pre-crisis period and expressed in standard deviations from pre-crisis conditions.	Osina (2019)
EPU_EU	Economic Policy Uncertainty index for Europe. Same construction as EPU_US, implemented on European newspapers and aggregated across European countries.	Baker et al. (2016)

Risk Factors Universe (Literature-Based, Reconstructed Series) X

EPU_US	Economic Policy Uncertainty index for the United States. Newspaper-based index constructed from the normalized frequency of articles that jointly reference the economy, uncertainty, and policy-related terms.	Baker et al. (2016)
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Panel F: Interest Rate Factors

Δ SLOPE_EU	Same construction as Δ SLOPE_US, computed using the 10-year German government bond yield minus the 3-month German government bill yield.	Ferson and Harvey (1991)
Δ SLOPE_US	First difference of the U.S. yield-curve slope, defined as the 10-year Treasury yield minus the 3-month Treasury bill yield.	Ferson and Harvey (1991)
EURUSD_3M_IV	First difference of the 3-month at-the-money implied volatility on the EUR/USD exchange rate.	Market indicator
SS2Y	Difference between the 2-year EUR fixed swap par rate and the 2-year German Schatz yield.	Collin-Dufresne et al. (2001)
SS5Y	Difference between the 5-year EUR fixed swap par rate and the 5-year German Bobl yield.	Collin-Dufresne et al. (2001)

Risk Factors Universe (Literature-Based, Reconstructed Series) XI

SS10Y	Difference between the 10-year EUR fixed swap par rate and the 10-year German Bund yield.	Collin-Dufresne et al. (2001)
TERM_EU	Same construction as TERM_US, computed using long-term German Bund total returns in excess of the 1-month German government bill return.	Fama and French (1993)
TERM_US	Bond term structure factor. Excess total return on long-term U.S. government bonds over the 1-month U.S. Treasury bill return.	Fama and French (1993)
YSP_EU	Yield-curve slope factor. Difference between the 5-year and 1-year German government bond yields.	Koijen et al. (2017)
YSP_US	Yield-curve slope factor. Difference between the 5-year and 1-year U.S. Treasury yields.	Koijen et al. (2017)

After reconstructing a comprehensive universe of 70+ risk factors, the next step is to identify a *small and interpretable set of controls* that best spans each arbitrage strategy return series.

Objective:

- identify the key factors that statistically span strategy returns
- test whether alpha survives after proper risk adjustment
- avoid overfitting in a high-dimensional setting

Because the candidate factor set is large relative to sample length, we rely on a disciplined **regularized factor-selection procedure**.

Why Adaptive Elastic Net? (Zou and Zhang, 2009)

With more than 70 candidate risk factors, standard OLS is not feasible:

- multicollinearity across controls
- overfitting in finite samples
- unstable estimates and unreliable alpha inference

We model each strategy return as:

$$r_t = \alpha + \beta' f_t + \varepsilon_t,$$

where f_t is a high-dimensional vector of candidate risk factors. **Adaptive Elastic Net** (Zou and Zhang, 2009) provides:

- shrinkage (stability)
- automatic variable selection (sparsity)
- interpretable factor benchmark (few selected controls)

Elastic Net estimates β by solving:

$$\hat{\beta}^{EN} = \arg \min_{\beta} \left\{ \frac{1}{T} \sum_{t=1}^T (r_t - \alpha - \beta' f_t)^2 + \lambda_2 \|\beta\|_2^2 + \lambda_1 \|\beta\|_1 \right\}.$$

- $\|\beta\|_1$ selects factors (LASSO effect)
- $\|\beta\|_2$ stabilizes correlated predictors (ridge effect)
- tuning parameters (λ_1, λ_2) chosen via a BIC-type criterion.

This yields a first sparse benchmark spanning the strategy.

Elastic Net is used as a first step to construct adaptive weights:

$$w_j = \left| \hat{\beta}_j^{EN} \right|^{-\gamma}$$

The final Adaptive Elastic Net solves:

$$\hat{\beta}^{AEN} = \arg \min_{\beta} \left\{ \frac{1}{T} \sum_{t=1}^T (r_t - \alpha - \beta' f_t)^2 + \lambda_2 \|\beta\|_2^2 + \lambda_1 \sum_{j=1}^p w_j |\beta_j| \right\}.$$

Interpretation:

- important factors are penalized less
- irrelevant factors are shrunk to zero
- output: a small set of 4–5 selected controls per strategy

A single Elastic Net run may select factors that are sample-specific.

We therefore apply **stability selection**:

- resample the data using moving-block bootstrap (6-month blocks),
- repeat the full Adaptive Elastic Net procedure,
- keep only factors consistently selected across replications.

Final benchmark:

$$S_i = \{\text{top 4–5 factors with highest selection frequency}\}.$$

This delivers economically stable risk controls.

After selecting a stable factor set S_i , we estimate:

$$r_{i,t} = \alpha_i + \beta_i' f_{S_i,t} + \varepsilon_{i,t}.$$

Key issue: factor selection creates post-selection bias.

Solution: **selection-inside bootstrap**

- resample returns and factors in blocks,
- re-run factor selection inside each draw,
- obtain confidence intervals for $\hat{\alpha}_i$.

This ensures alpha significance is robust to model uncertainty.

AEN Stable-Factor Model: BTP Italia

Variable	Coefficient	<i>t</i> -stat	<i>p</i> -value
α	+1.62***	3.71	0.00
ILLIQ	+0.88***	3.81	0.00
UMD _{EU}	+0.05***	4.49	0.00
SMB _{EU}	-0.05**	-2.43	0.02
BAB _{US}	-0.04**	-2.49	0.01
BTP _{BUND}	+0.00**	2.07	0.04
TERM _{EU}	-0.02**	-2.38	0.02
CREDIT _{EU}	+0.04	0.09	0.93
SS10Y	-0.02***	-3.26	0.00

Model fit

T (obs)	158
k (factors)	8
R^2	0.2858
R^2_{adj}	0.2474
Durbin–Watson	2.0553
F -stat	7.4518
$p(F)$	0.0000

(α) annualized %.
*** $p < 1\%$, ** $p < 5\%$,
* $p < 10\%$.

- **ILLIQ**: Amihud illiquidity shock ($\Delta \text{ret} - \text{volume}$).
- **UMD_{EU}**: European equity momentum factor (winners – losers).
- **SMB_{EU}**: European equity size factor (small – big caps).
- **BAB_{US}**: US betting-against-beta factor (low – high beta).
- **BTP_{BUND}**: Italian sovereign risk (10Y BTP – Bund yield spread).
- **TERM_{EU}**: Euro term premium (long-term Bund excess return over German bill).
- **CREDIT_{EU}**: Euro credit spread factor (BBB – AAA bond yield).
- **SS10Y**: 10Y EUR swap spread (par swap rate – 10Y Bund yield).

Amihud and Mendelson (2015)

Carhart (1997)

Fama and French (2017)

Frazzini and Pedersen (2014)

Fausch and Sigonius (2018)

Fama and French (1993)

Fama and French (1989)

Collin-Dufresne et al. (2001)

AEN Stable-Factor Model: CDS–Bond Basis

Variable	Coefficient	<i>t</i> -stat	<i>p</i> -value
α	+5.82***	5.40	0.00
EBP	+0.27**	2.23	0.03
CRED _{SPR_US}	-0.59**	-2.04	0.04
TERM _{US}	+0.08***	2.71	0.01
RI _{EU}	+0.09	1.22	0.22
PB _{EU_CDS_1Y}	+0.02	1.50	0.13
SMB _{EU}	+0.05	1.41	0.16
SS10Y	+0.02**	2.34	0.02
DEF _{US}	+0.11***	3.83	0.00

Model fit	
<i>T</i> (obs)	201
<i>k</i> (factors)	8
<i>R</i> ²	0.2518
<i>R</i> _{adj} ²	0.2206
Durbin–Watson	1.7658
<i>F</i> -stat	8.0772
<i>p</i> (<i>F</i>)	0.0000

(α) annualized %.
 *** $p < 1\%$, ** $p < 5\%$,
 * $p < 10\%$.

- EBP: Excess bond premium: corporate spread net of expected default.
- CRED_{SPR_US}: US credit spread factor (Moody's BAA – AAA yield).
- TERM_{US}: US term premium (long-term gov. bond excess return over T-bill).
- RI_{EU}: Euro investment-grade industrial bond return factor.
- PB_{EU_CDS_1Y}: European prime broker 1Y CDS spread (counterparty risk).
- SMB_{EU}: European equity size factor (small – big caps).
- SS10Y: 10Y EUR swap spread (par swap rate – 10Y Bund yield).
- DEF_{US}: US default factor (corporate – gov. long-term bond return).

Gilchrist and Zakrajsek (2012)
Fama and French (1989)
Fama and French (1993)
Collin-Dufresne et al. (2001)
Klaus and Rzepkowski (2009)
Fama and French (2017)
Collin-Dufresne et al. (2001)
Fama and French (1993)

AEN Stable-Factor Model: iTraxx Combined

Variable	Coefficient	t-stat	p-value
α	+1.86***	2.59	0.01
EP _{SVIX_1M}	+3.08	1.15	0.25
LIBOR _{REPO_SHOCK}	+0.55***	2.87	0.00
SILLIQ	+0.01*	1.94	0.05
EP _{SVIX_3M}	+1.95	1.06	0.29
UF	-1.26	-0.86	0.39
TED _{SHOCK_EU}	+0.56**	2.41	0.02
R10 _{EU}	+0.17	1.25	0.21

Model fit

T (obs)	237
k (factors)	7
R^2	0.2070
R^2_{adj}	0.1828
Durbin–Watson	1.8400
F -stat	8.5403
$p(F)$	0.0000

(α) annualized %.
 *** $p < 1\%$, ** $p < 5\%$,
 * $p < 10\%$.

- EP_{SVIX_1M}: Option-implied equity premium via SVIX, 1-month horizon. *Martin (2017)*
- LIBOR_{REPO_SHOCK}: Dollar funding liquidity shock (LIBOR – GC repo, AR(2) res.). *Asness et al. (2013)*
- SILLIQ: Stock-market illiquidity innovation (AR(3) residual). *Acharya et al. (2013)*
- EP_{SVIX_3M}: Option-implied equity premium via SVIX, 3-month horizon. *Martin (2017)*
- UF: Change in financial uncertainty index. *Ludvigson et al. (2021)*
- TED_{SHOCK_EU}: Euro interbank funding shock (Euribor – German Bill, AR(2) res.). *Asness et al. (2013)*
- R10_{EU}: Euro 10Y government bond return factor. *Cochrane and Piazzesi (2005)*